3 **(a)** Fig. 3.1 shows two chambers, X and Y, connected by a small pipe which is fitted with a valve. Both chambers are filled with ideal gas and the valve was initially closed. The volume of chambers X and Y are 2.5 m^3 and 4.0 m^3 respectively. Chambers X and Y are held at temperatures of 450 K and 300 K respectively.

The valve is then opened and a state of equilibrium is reached with the temperatures in each chamber remaining unchanged.

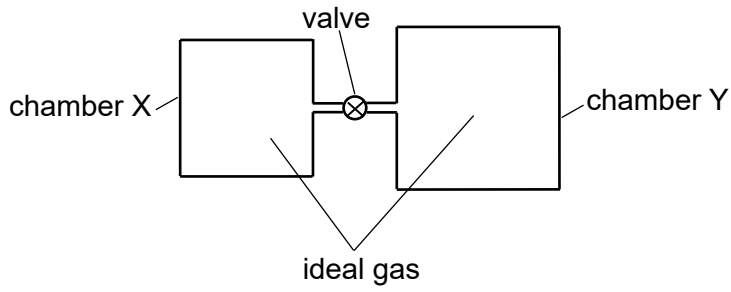

Fig. 3.1

(i) Determine the number of moles of ideal gas in chamber X, given that the number of moles of ideal gas in chamber Y is 1.2 after equilibrium has been reached.

number of moles of ideal gas in X = ... [2]

(ii) Calculate the pressure in both chambers after equilibrium has been reached.

pressure = ... Pa [1]

(iii) The valve is now closed and a pump is used to remove some ideal gas from chamber Y.

State and explain, using the kinetic theory of gas, why the pressure in chamber Y decreases, assuming no change in its temperature.

.....

.....

.....

..... [1]

(b) A system of a fixed mass of ideal gas undergoes a cycle of changes. Fig. 3.2 shows the variation with volume V of the pressure p of the ideal gas as it undergoes the cycle ABCA.

Process A to B is isothermal, process B to C is isovolumetric and process C to A is adiabatic where there is no heat transfer into or out of the system of ideal gas.

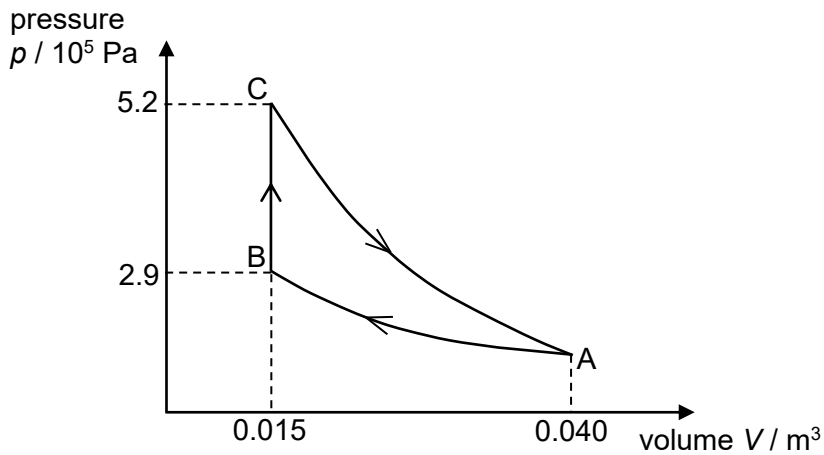

Fig. 3.2

Given that the mass of the ideal gas is 0.060 kg, calculate

(i) the pressure of the ideal gas in state A,

pressure = ... Pa [2]

(ii) the root-mean-square (r.m.s.) speed of the ideal gas molecules in state A,

r.m.s. speed = ... m s^{-1} [2]

(iii) the thermal energy absorbed by the system of ideal gas from B to C.

thermal energy = ... J [3]

(iv) Suggest one other way in which the adiabatic process C to A could be achieved in practice, other than by thermally insulating the system of gas.

.....

..... [1]